Samin Hussain

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Education

University of Waterloo Waterloo, ON
Honors Mechanical Engineering, Co-op *Sept. 2026 – May 2031*

Skills

Technical: Microsoft Office (Word, Excel, PowerPoint), SolidWorks, Fusion 360, HTML, Python, CSS
CAD & Design: Fusion 360, BambuStudio, AutoCAD, SolidWorks FEA/Design, OpenRocket, Machine Shop
Professional: Organization, Time management, Teamwork, Communication, Leadership, Fast learner

Projects

1-DOF Mechanical Gripper | *Multi-Actuator Design* August 2026

- Designed a single-DOF mechanical gripper driven by four commercial actuators, selected against voltage, mass, and torque constraints to lift objects up to 15 lb
- Engineered structural frames and supporting mechanisms that interface the actuators and translate their output into desired ROM in **SolidWORKS**
- Applied standard fastener and hole-sizing conventions and **CAD best practices** across iterative design to ensure manufacturable, assemblable joints

Waterloo Rocketry | *Flight Dynamics Analyst/Machinist* August 2026

- Conducting retrospective analysis of the Polaris rocket, examining changes in mass, geometry, and system architecture across the 2026 design cycle
- Recreated the rocket's flight using simulations in **OpenRocket**, comparing predicted altitude and landing location against flight results, utilizing **Microsoft Excel**
- Analyzed simulation-to-flight discrepancies to identify potential sources of model error and opportunities to improve future flight simulations and design decisions in the 2027 model
- Applied principles of systems engineering, **data analysis**, modeling, and iterative engineering design to evaluate real-world rocket performance

Skills-Based Volunteering

Volunteer — Youth Central & Islamic Information Society of Calgary March 2019 – Present
Calgary, AB

- Took leadership roles in community events and programs, including golf tournaments, charity pickleball, and national track and field, and supported national conferences, charity dinners, and youth summits with **150+** hours of combined service
- Assisted with food services at Calgary homeless shelters, including the Salvation Army, Inn from the Cold, and Calgary Drop-In Centre, and supervised mosque operations while maintaining composure in high-pressure situations
- Provided weekly one-on-one Quran teaching, tutored children in coding, and helped seniors with technology, strengthening communication and teaching skills

Activities & Leadership

Co-Founder – Muslim Youth Club, North Trail High School Sept. 2023 – June 2026

- Collaborated with various schools, including the University of Calgary, to help raise approximately **$63,000** for orphans around the world
- Initiated the creation of the club to support an inclusive environment where Muslim students could build community and practice their religion

U19 North Calgary Football Club – Calgary Minor Soccer Association Feb. 2025 – July 2026

- Competed at the competitive Tier 1 level while balancing a full academic course load
- Developed cooperation, communication, and teamwork skills both on and off the pitch